Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years writing **Python** and **SQL** for data pipelines, schema design, and access-controlled data, adding **validation checks**, automating repeat manual work, and supporting the users of production tools.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed relational schemas with **row-level security** for 500+ users, controlling who could see and edit each tenant's data, and tuned indexes that cut dashboard latency from 1.2s to 350ms.
- Built **Python** and **SQL** data pipelines with incremental loads, **validation and reconciliation checks**, and CI/CD-backed releases across PostgreSQL and SQL Server, sustaining 99%+ reliability over 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Supported 150+ instructors on a production grading platform handling 10K+ daily requests, **debugging** OCR and model failures from logs and adding fallbacks that kept Flask REST APIs stable at peak load.
- **Automated** a manual grading workflow with LLM prompt chains for rubrics, OCR, and feedback plus per-call usage logging in MySQL, cutting instructor feedback turnaround 60% across 1K+ files per batch.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python pipelines turning 100K+ raw trajectory logs and map metadata into **research datasets** with Hydra configs, and tracked experiments in Weights & Biases, improving long-horizon planning 18%.
- Ran controlled comparisons of a new attention encoder against three baseline encoders in closed-loop simulation, documenting **failure modes** such as off-road trajectories and improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Automated** a nightly Jenkins job that pulled builds, installed them on test VMs, ran 20+ PyTest and Mockito suites, and emailed CSV reports, retiring manual test runs across Windows, Linux, and macOS.
- Redesigned AWS Glue ETL that normalized JSON, CSV, and Parquet from multiple sources, and **debugged** API integrations from failure logs, improving pipeline efficiency 68% and response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a role-based **access control** service and Python REST microservices for a multi-site platform, modeling Azure SQL tables for users, resources, and allocations while syncing 2M+ records across locations.
- Shipped schema and code changes through Jenkins **CI/CD** with linting, unit, integration, and end-to-end tests and approval-gated staging releases, reaching 95%+ successful build validation and halving deployment time.

Skills

Languages & Process: Python, SQL, Bash, Shell, Java, JavaScript, TypeScript, Agile, User Support
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, Schema Design, Data Modeling, Data Validation, ETL Pipelines
Backend & Automation: REST APIs, FastAPI, Flask, Microservices, Jenkins, GitHub Actions, CI/CD, Docker
AI & Research: LLM Systems, Prompt Engineering, PyTorch, Hydra, Weights & Biases, Experiment Tracking, AWS, Linux
Quality & Support: Role-Based Access Control, Row-Level Security, PyTest, Debugging, Log Analysis, Documentation, Git

Projects

Lab Metadata Entry Validator | *Python, SQL, REST APIs, PyTest, GitHub Actions*

- Designed **data-entry checks** that validate sample and experiment records against required fields, controlled vocabularies, ID formats, and plate positions, with rule changes tested in PyTest and shipped through **CI**.

Experiment Metadata Query CLI | *Python, PostgreSQL, REST APIs, Docker, LLM APIs*

- Designed a **command-line tool** and scheduled sync job that join sample, run, and sequencing metadata from separate systems into one **queryable** store, plus an LLM agent that drafts answers to repeat requests.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker*

- Extended a **SQL** workspace supporting 10+ databases with schema exploration, structure synchronization across environments, and AI-generated queries reaching 82% accuracy, set up with Docker and Maven for local development.